**ROLL NO:** | | | | | | | | | | |

**(To fill the roll no is compulsory)**

INVERTIS UNIVERSITY BAREILLY

# EVEN SEMESTER EXAMINATION - 2024-25

## B.TECH (CSE)

## Semester: IV

## SUBJECT CODE: BCS401

## COMPUTER ORGANIZATIONAND AND ARCHITCTURE

**Time: 03:00 Hrs.** **Maximum Marks: 50**

**NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.**

### SECTION – A

**NOTE: Attempt all parts of the following question.** **(10 Marks)**

**Q. 1. Define the following terms:** **(1 × 5 = 5 Marks)**

**(i)** Bus System.
**(ii)** Immediate addressing.
**(iii)** Polling Method.
**(iv)** Register Transfer language.
**(v)** Micro Operations.

**Q. 2. Answer the following questions:** **(1 × 5 = 5 Marks)**

**(i)** Instruction cycle.
**(ii)** Speedup.
**(iii)** Pipelining.
**(iv)** Virtual Memory.
**(v)** Perform $43_{10}$-$25_{8}$.

### SECTION – B

**NOTE:Attempt all questions.** **(5×2×2 = 20 Marks)**

**Q.3.(i)(a)** Explain about bus and bus arbitration? Draw and Discuss between various types of bus arbitration.

**OR**

(**b)** What is Interrupts? Describe types of interrupts?



**(ii)(a)** What is the need of addressing modes? Discuss about different types of addressing mode with example.

**OR**

**(b)** What do you mean by Modes of Data Transfer? Explain in brief?

**Q.4.(i) (a)** Describe about the organization of stack . Differentiate between Memory stack and Register stack in detail.

**OR**

(**b)** Perform arithmetic operation using 1's and 2's Complement.
**(1).** 442 - 212 **(2).** 1252 - 642

**(ii)(a)** Evaluate the arithmetic statement X = (A*B) - (C/D) using a general register computer with three and two address instruction format.

**OR**

(**b)**What do you mean by DMA? Explain in brief with neat diagram.

### SECTION – C

**NOTE: Attempt all questions.** **(10 × 2 = 20 Marks)**

**Q.5.(a)** Represent the number (-1035.325)10 in single precision and double precision format using IEEE standard for Explain IEEE standard for floating point.

**OR**

**(b)** Draw and design about memory hierarchy. Discuss about different kinds of memory with neat diagram.

**Q.6.(a)** What do you mean by Micro Operation ? How many types of operation? Explain in brief.

**OR**

**(b)** Explain Booths Algorithms? Multiply 910 by - 610 using Booth's Algorithms?